

California State University Long Beach

Team Auger

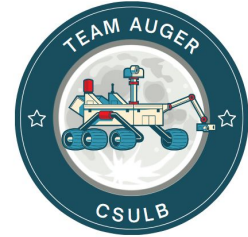


System Requirements Review

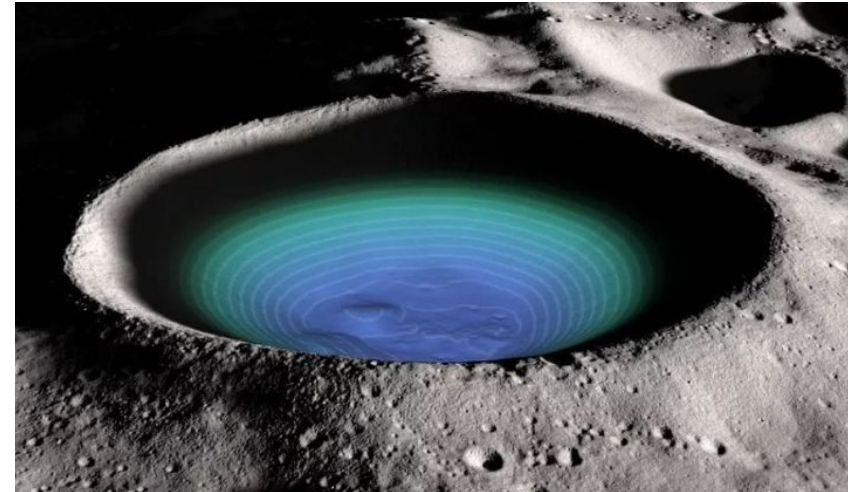
September 04, 2023



Agenda



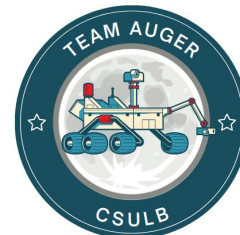
- Team Introductions
- Proposed Project Overview
 - Proposed Project Summary
 - Proposed Mission Profile
- Major Project Level 1 Requirements
 - Proposed Overall Project V&V plan
 - Proposed Design Approach
- Subsystem Overview
 - Proposed Level 2 Requirements & Proposed Verification Method
 - Proposed Design
 - Proposed Design Interface
 - Planned Subsystem Trade Studies
- Conclusion
- Questions



<https://www.indiatimes.com/technology/science-and-future/nasa-moon-water-discovery-526143.html>



Team Introduction



Alex Contreras

Team Lead &
Heat Transfer
Lead



Experience in
Power Budgets,
Subsystem
Integration,
Design Reviews

Zachary Tep
Avionics Lead



Experience in
structure,
manufacturing
and propulsion.

**Alondra
Barba**
Structures Lead



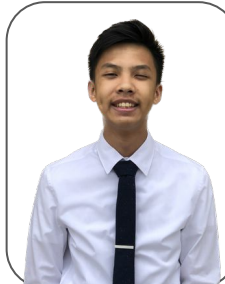
Experience in
structure,
design, and
manufacturing

David Guerra
GNC Lead



Experience in
structures, fluids
mechanics and
aerodynamics.

Erick Nguyen
Mechanical
Design lead



Experience in
mechanical
design and
structural
analysis

Bryan Ramos
Telecommuni-
cations lead



Experienced in
structural analysis,
fluids and
Stability/control

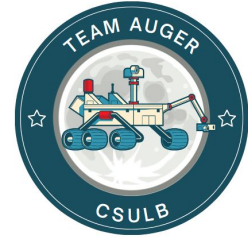
**Damon
Nguyen**
Power Lead



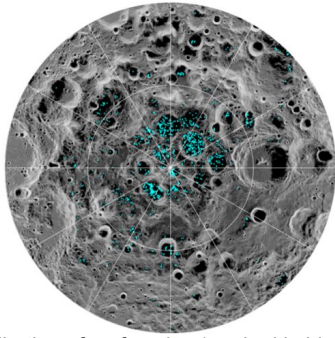
Experience in
thermodynamic,
structures,
aerodynamics,
and fluids.



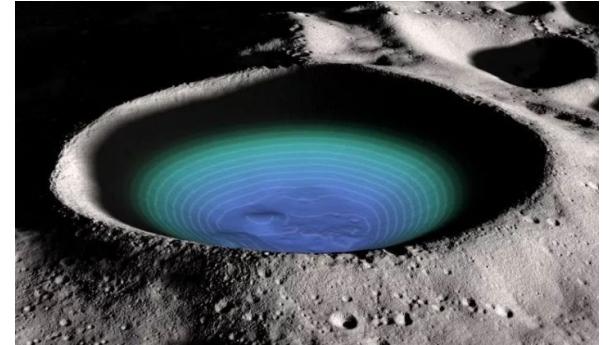
Project Overview



To design an autonomous lunar rover to demonstrate its ability to extract water ice on the lunar surface. The autonomous rover will be engineered to identify areas in the permanently shadowed region with potential water ice deposits. It will then be able to collect and store water ice samples in a pressurized storage tank to avoid sublimation. The rover will be able to withstand the moon's extreme cold temperatures. The rover will aid in the goal of establishing a habitable environment on the lunar surface by demonstrating its ability to extract and store water ice.



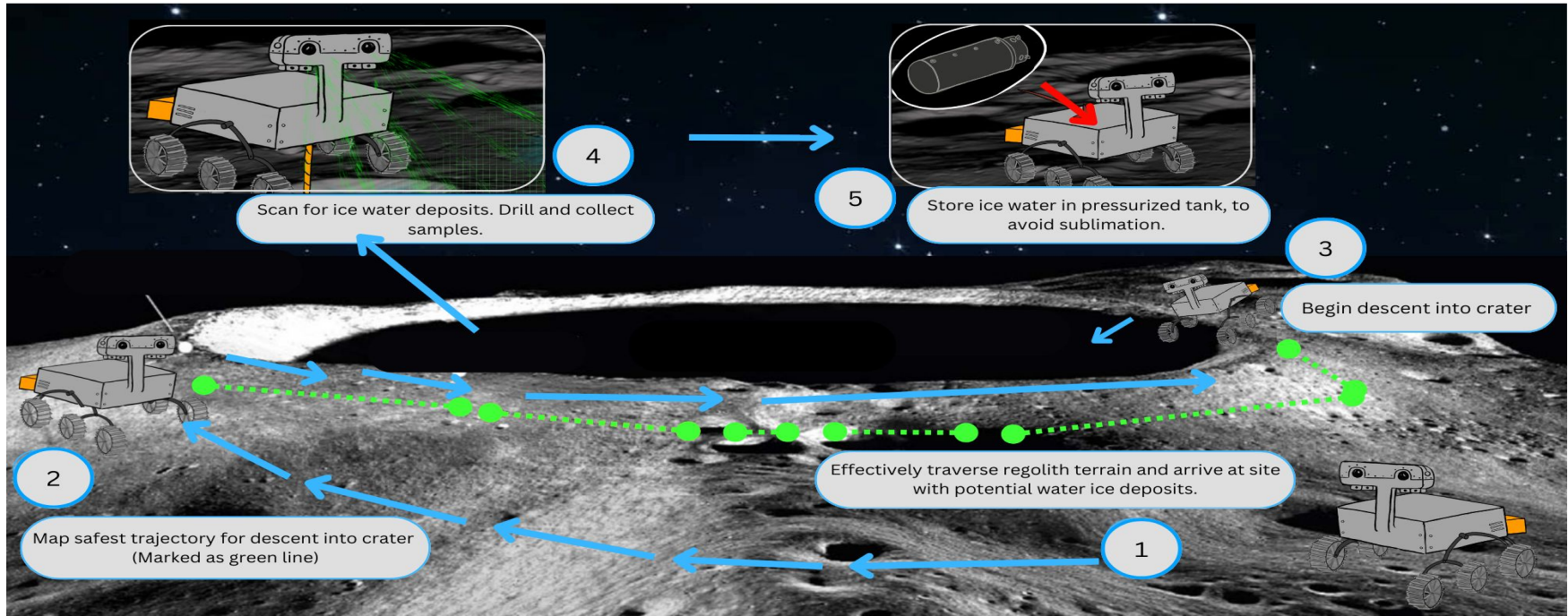
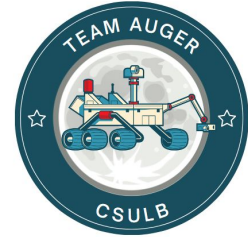
The distribution of surface ice (marked in blue) at the Moon's south poles. For more information [visit](https://www.indiatimes.com/technology/science-and-future/nasa-moon-water-discovery-526143.html).



<https://www.indiatimes.com/technology/science-and-future/nasa-moon-water-discovery-526143.html>

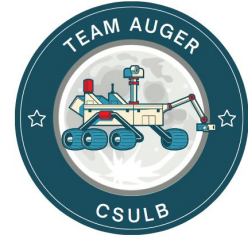


Project Overview





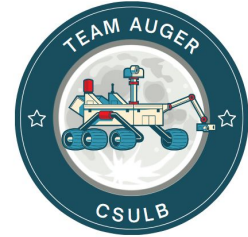
Level I Requirements



Requirement #	Requirement	Verification Method	Validation Statement
R 1.1	The rover shall autonomously identify areas in the permanently shadowed region with water ice deposits with an 80%±5 accuracy (TBR).	Test	Be able to search for water ice deposits
R 1.2	The rover shall collect a minimum of 20 (TBR) samples of water ice over the entire mission life cycle.	Test	Design requirement
R 1.3	Each water ice sample shall be 100g ±5g (TBD) .	Test	Design Requirement
R 1.4	The rover shall be operational for a minimum of 50 days (TBD).	Test	Design requirement
R 1.5	The rover shall travel a minimum distance of 10km (TBD).	Test	Design requirement



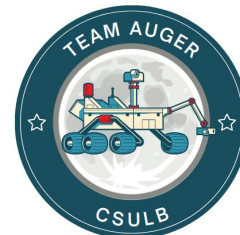
Level I Trade Studies



Crater Selection			ENTER SCORE ↓	Cabeus	Shackleton	Nobile	Shoemaker
CRITERIA	MANDATORY Y=1 / N=0	WEIGHT	SCALE				
Water Ice Possibility	1	30%	3 = High Possibility 1 = Low Possibility	3	3	2	2
Existing Data/Knowledge	1	20%	3 = Most Researched 1 = Least Researched	2	3	3	3
Crater Wall Slope	1	25%	3 = Least Incline 1 = Most Incline	3	1	2	3
Landing Site Suitability	1	25%	3 = Most Suitable 1 = Least Suitable	2	2	3	2
WEIGHTED TOTALS		100%		85%	75%	82%	82%



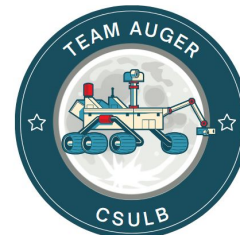
Level I Trade Studies



Number of Wheels			ENTER SCORE ↓	4-Wheel Configuration	6-Wheel Configuration (Equally Distributed)	6-Wheel Configuration (2+4 Layout)
CRITERIA	MANDATORY Y=1 / N=0	WEIGHT	SCALE			
Mobility	1	25%	3 = Most mobile 1 = Least mobile	2	2	1
Stability	1	25%	3 = Most stable 1 = Least stable	2	3	1
Weight Distribution	1	25%	3 = Least heavy 1 = Most heavy	2	3	1
Power consumption	1	25%	3= Least consumption 1= Most consumption	3	2	2
WEIGHTED TOTALS		100%		74.5%	83%	44%



Level I Trade Studies



Wheel Drive			ENTER SCORE ↓	Independent Wheel Drive	Front 2-Wheel Drive	4-Wheel Drive
CRITERIA	MANDATORY Y=1 / N=0	WEIGHT	SCALE			
Power Efficiency	1	40%	3 = Most efficient 1 = Least efficient	2	3	1
Traction	1	25%	3 = Most traction 1 = Least traction	3	1	2
Maneuverability	1	20%	3 = Most maneuverable 1 = Least maneuverable	3	1	2
Speed	1	15%	3= Most Speed 1= Least Speed	3	1	2
WEIGHTED TOTALS		100%		86.4 %	59.8 %	52.8%



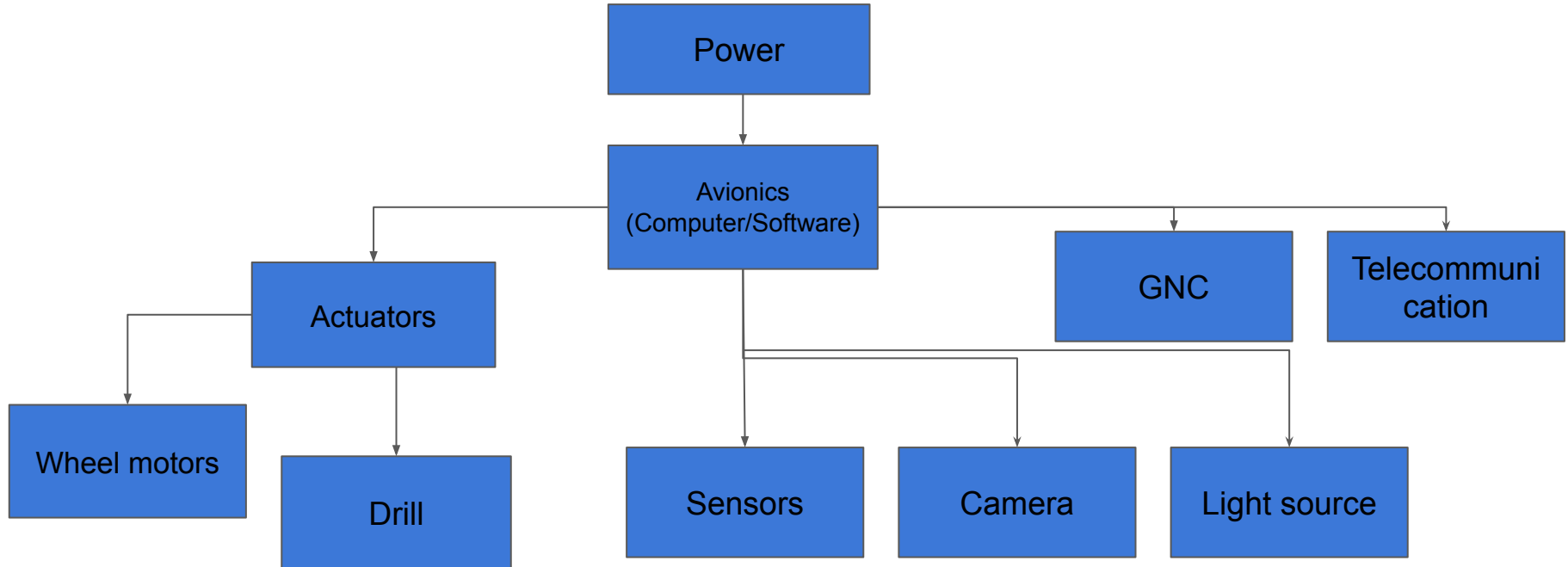
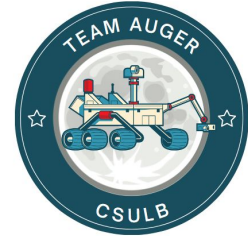
Proposed Avionics Requirements



Requirement #	Requirement Description	Verification Method	Validation Statement
R 2.1.1	The rover shall scan areas of $5 \pm 2 \text{ m}^2$ (TBR) for presence of water ice.	Test	Locating water ice is the first step towards collecting it
R 2.1.2	The rover shall differentiate water ice from lunar regolith with an accuracy of $90\% \pm 5$. (TBR)	Test	To prevent collecting the wrong samples
R 2.1.3	The rover shall verify the amount of collected water ice every 5 ± 0.2 seconds. (TBR)	Test	To prevent the overcollection of water ice
R 2.1.4	The rover shall have a depth sensor with an accuracy of $90\% \pm 5$. (TBR)	Test	To ensure the drill does not go past the required depth



Avionics Proposed Interfaces





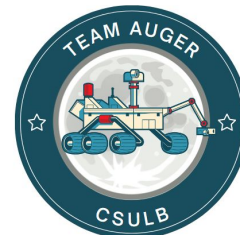
Avionics Trade Study



Water Ice Detection System			ENTER SCORE ↓	Neutron Spectrometer	Thermal Infrared Spectrometer	X-Ray Fluorescence Spectrometer
CRITERIA	MANDATORY Y=1 / N=0	WEIGHT	SCALE			
Sensitivity to water ice	1	35%	3 = Most sensitive 1 = Least sensitive	3	2	1
Area Coverage	1	25%	3 = Most range 1 = Least range	2	2	2
Size/Weight	1	15%	3 = Least heavy 1 = Most heavy	3	1	3
Depth sensing	1	25%	3= Most depth 1= Least depth	3	1	1
WEIGHTED TOTALS		100%		92%	50%	58%



Avionics Trade Study



Camera Type			ENTER SCORE ↓	Mastcam	Navcam	Hazcam
CRITERIA	MANDATORY Y=1 / N=0	WEIGHT	SCALE			
Resolution	1	30%	3 = Most resolution 1 = Least resolution	3	2	1
Durability	1	30%	3 = Most durable 1 = Least durable	2	3	3
Data Storage	1	20%	3 = Most storage 1 = Least storage	3	3	1
Light Sensitivity	1	20%	3= Least heavy 1= Most heavy	2	3	1
WEIGHTED TOTALS		100%		83%	92%	50%



Avionics Trade Study



Water Ice Measuring System			ENTER SCORE ↓	Level Sensor	Pressure Sensor	Load Cell
CRITERIA	MANDATORY Y=1 / N=0	WEIGHT	SCALE			
Data Accuracy	1	35%	3 = Most accurate 1 = Least accurate	3	2	3
Feedback Control	1	30%	3 = Most feedback 1 = Least feedback	2	2	3
Size/Weight	1	15%	3 = Least heavy 1 = Most heavy	3	2	2
Durability	1	20%	3= Least expensive 1= Most expensive	2	3	3
WEIGHTED TOTALS		100%		83%	75%	91%



Avionics Trade Study



Light Source			ENTER SCORE ↓	LED Light Source	Halogen Light Source	Xenon Light Source
CRITERIA	MANDATORY Y=1 / N=0	WEIGHT	SCALE			
Brightness	1	30%	3 = Brightest 1 = Least brightest	2	2	3
Durability	1	30%	3 = Most durable 1 = Least durable	3	2	2
Lifetime	1	25%	3 = Most lifetime 1 = Least lifetime	2	2	3
Energy Efficiency	0	15%	3= Most energy efficient 1= Least energy efficient	3	1	1
WEIGHTED TOTALS		100%		83%	58%	75%



Proposed Structure Requirements

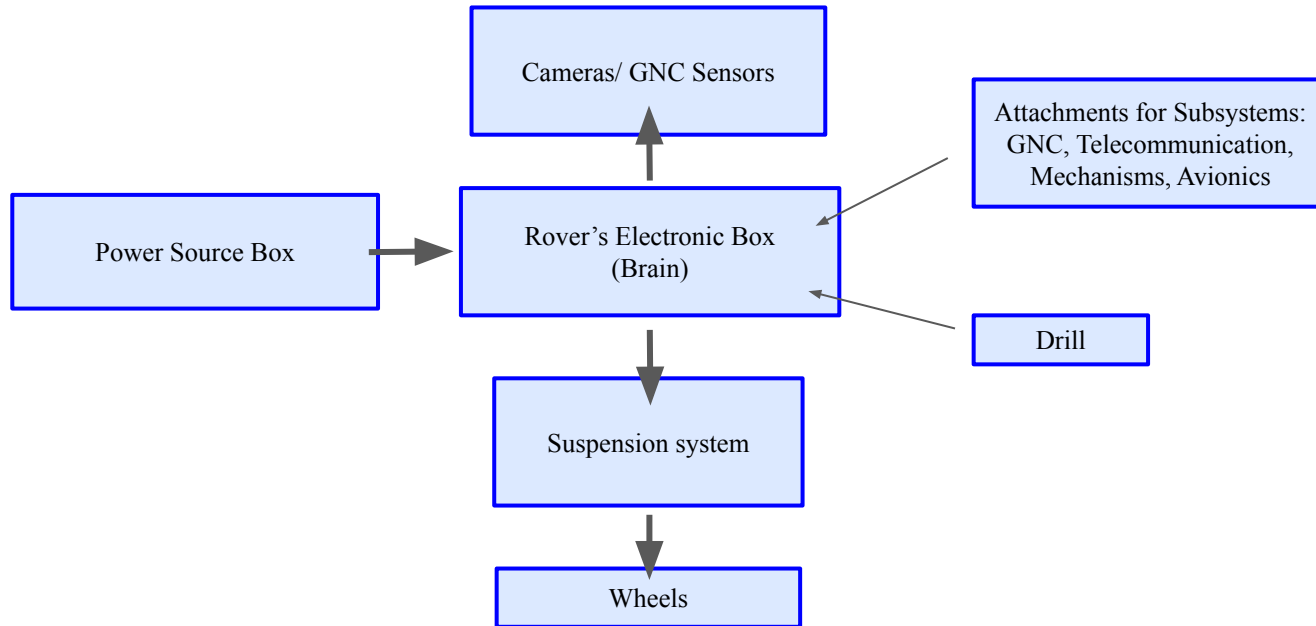
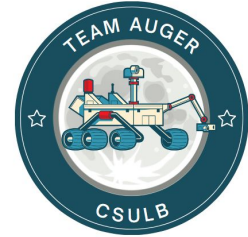


Purpose: The autonomous rover will be capable of traversing the lunar regolith with maximum stability and efficiency.

Requirement #	Requirement Description	Verification Method	Validation Statement
R 2.2.1	The structure shall provide a subsystem internal volume of 3.5 cubic meters $\pm$ 0.1 cubic meters. (TBR)	Test and Analyze	Volumetric space needed for additional subsystems.
R 2.2.2	The structure shall support maximum payload mass of 800 kg. (TBR)	Test and Analyze	Rover shall be able to traverse irregular lunar terrain and heavily created regions.
R 2.2.3	The structure shall protect against maximum radial absorption of 10.5 μ Gy/hr. (TBR)	Test and Analyze	The structure material must be able to withstand radiation on the lunar surface.
R 2.2.4	The structure material shall have maximum outgassing rate of 5.0×10^{-7} mbarl/s*cm ² . (TBR)	Test and Analyze	Withstand the weight of science instrument and sample payload.



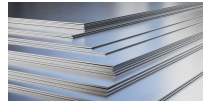
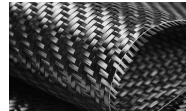
Structure Subsystem Proposed Interfaces





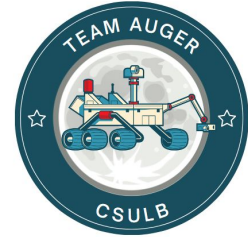
Structure Subsystem Trade Studies

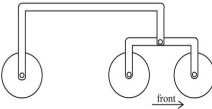
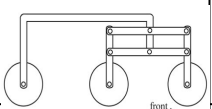
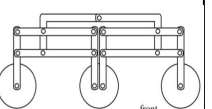


<u>Trade Space:</u> Material for Chassis Frame			ENTER SCORE ↓	Titanium Aluminum Alloy	Carbon Fiber Composites	Aluminum Nickel Alloy
CRITERIA	MANDATORY Y=1 / N=0	WEIGHT		SCALE		
Environmental Conditions	1	35 %	3= Most Suitable 1= Least Suitable	3	0	2
Material Strength	1	30 %	3= Most Durable 1= Least Durable	2	3	1
Thermal Resistance	1	20 %	3= Most Resistant 1= Least Resistant	3	1	2
Mass	1	15 %	3= Most Weight 1= Least Weight	2	3	1
WEIGHTED TOTALS		100 %		84.70%	51.05%	61.05%



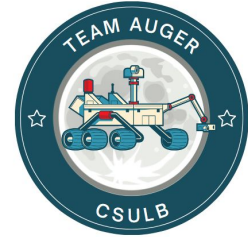
Structure Subsystem Trade Studies

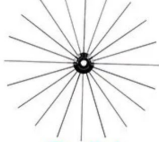


Trade Space: Suspension System Configuration			ENTER SCORE ↓			
CRITERIA	MANDATORY Y=1 / N=0	WEIGHT	SCALE			
Directional Stability	1	35 %	3= Most Stable 1= Least Stable	 2	 3	 3
Control	1	30 %	3= Most Control 1= Least Control	1	3	3
Road Isolation	1	25 %	3= Most Absorption 1= Least Absorption	1	3	2
Speed	0	10 %	3= Most Speed 1= Least Speed	1	1	1
WEIGHTED TOTALS		100 %		44.55 %	93.3 %	84.8 %



Structure Subsystem Trade Studies



<u>Trade Space:</u>		Spoke Wheel Lacing Patterns		ENTER SCORE ↓	Radial	2 Cross Lacing	4 Cross Lacing
CRITERIA	MANDATORY Y=1 / N=0	WEIGHT	SCALE				
Stress Bearing Capacity	1	40 %	3 = Most Efficient 1 = Least Efficient		2	2	3
Load Capacity	1	25 %	3 = Most Capable 1 = Least Capable		1	2	3
Stability	1	20 %	3 = Most Stable 1 = Least Stable		2	3	3
Mass	1	15 %	3 = Least Weight 1 = Most Weight		3	2	1
WEIGHTED TOTALS		100%			62.85 %	72.8 %	94.9 %



Proposed Mechanisms Requirements

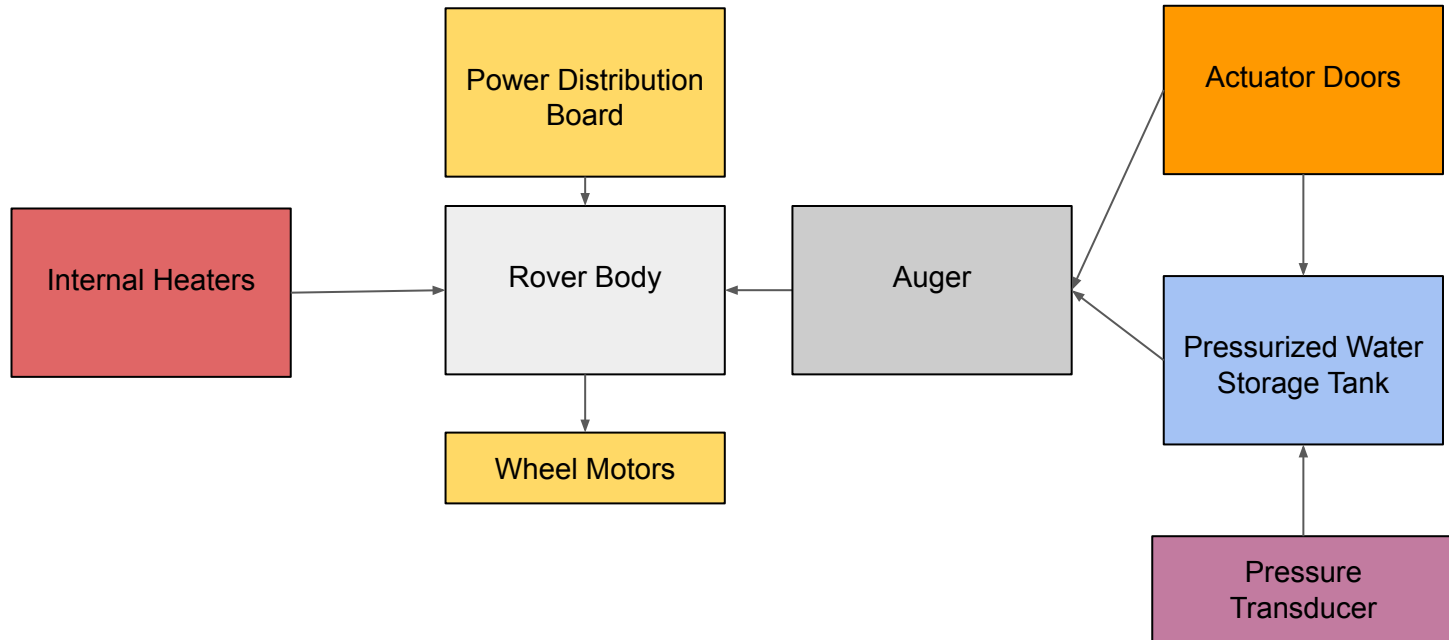
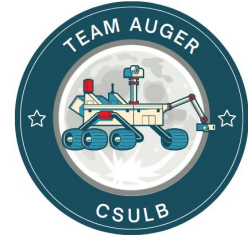


Purpose: The autonomous rover will be capable drilling into lunar regolith and extract ice water .

Requirement #	Requirement Description	Verification Method	Validation Statement
R 2.3.1	The rover shall retrieve $2000\text{g} \pm 200\text{g}$ of water ice within its chassis.	Test	Provide sufficient storage for water samples
R 2.3.2	The rover drilling mechanism shall maintain an internal pressure of $115\text{ kPa} \pm 5\text{ Pa}$. (TBR)	Test	Prevents sublimation caused by component generated heat
R 2.3.3	The rover drill shall achieve a drilling depth of at $1\text{m} \pm .5\text{m}$ (TBR)	Test	Provide depth to extract ice core samples beneath lunar surface
R 2.3.4	The rover shall drill $0.15\text{m} \pm .05\text{m}$ every two hours (TBR)	Test	Allows for efficient drilling



Mechanisms Subsystem Proposed Interfaces





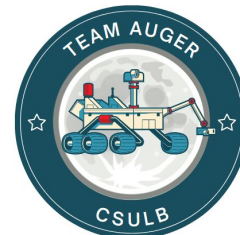
Mechanisms Subsystem Proposed Trade Studies



Wheel Motors			ENTER SCORE ↓	Brushless DC Motor	Brushed DC Motor	Wheel Hub Motors
CRITERIA	MANDATORY Y=1 / N=0	WEIGHT	SCALE			
Reliability	1	40%	3 = Most Reliable 1 = least Reliable	3	2	3
Power Efficiency	1	30%	3 = Most Efficient 1 = Least Efficient	3	2	3
Weight	1	20%	3 = Least heavy 1 = Most heavy	3	2	2
Cost	0	10%	3= Least Expensive 1= Most expensive	2	3	2
WEIGHTED TOTALS		100%		96.67%	69.99%	90%



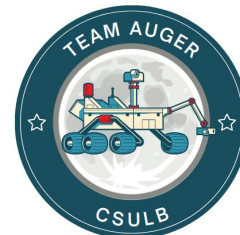
Mechanisms Subsystem Proposed Trade Studies



Door Actuator			ENTER SCORE ↓	Pneumatic Rotary Actuators	Electric Motor Actuators
CRITERIA	MANDATORY Y=1 / N=0	WEIGHT	SCALE		
Precision/Accuracy	1	50%	3 = most precise 1 = least precise	2	3
Integrate Power Source	1	20%	3 = Integrated 1 = Needs to be integrated	1	3
Size/Weight	1	20%	3 = Least heavy 1 = Most heavy	2	2
Cost	0	10%	3= Least Expensive 1= Most expensive	3	2
WEIGHTED TOTALS		100%		63.33%	90%



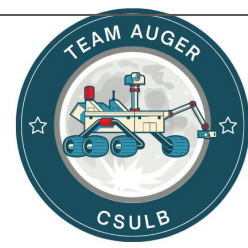
Mechanisms Subsystem Proposed Trade Studies



Pressure Transducers			ENTER SCORE ↓	Strain-Gauge Pressure Transducer	Capacitive Pressure Transducer	Piezoelectric
CRITERIA	MANDATORY Y=1 / N=0	WEIGHT	SCALE			
Temperature Sensitivity	1	35%	3 = least sensitive 1 = most sensitive	3	2	1
Accuracy/Reliability	1	35%	3 = Most suitable range 1 = Least suitable range	3	3	2
Size/Weight	1	20%	3 = Least heavy 1 = Most heavy	2	3	1
Cost	0	10%	3= Least Expensive 1= Most expensive	3	2	2
WEIGHTED TOTALS		100%		93.33%	85%	50%



Proposed GNC Requirements



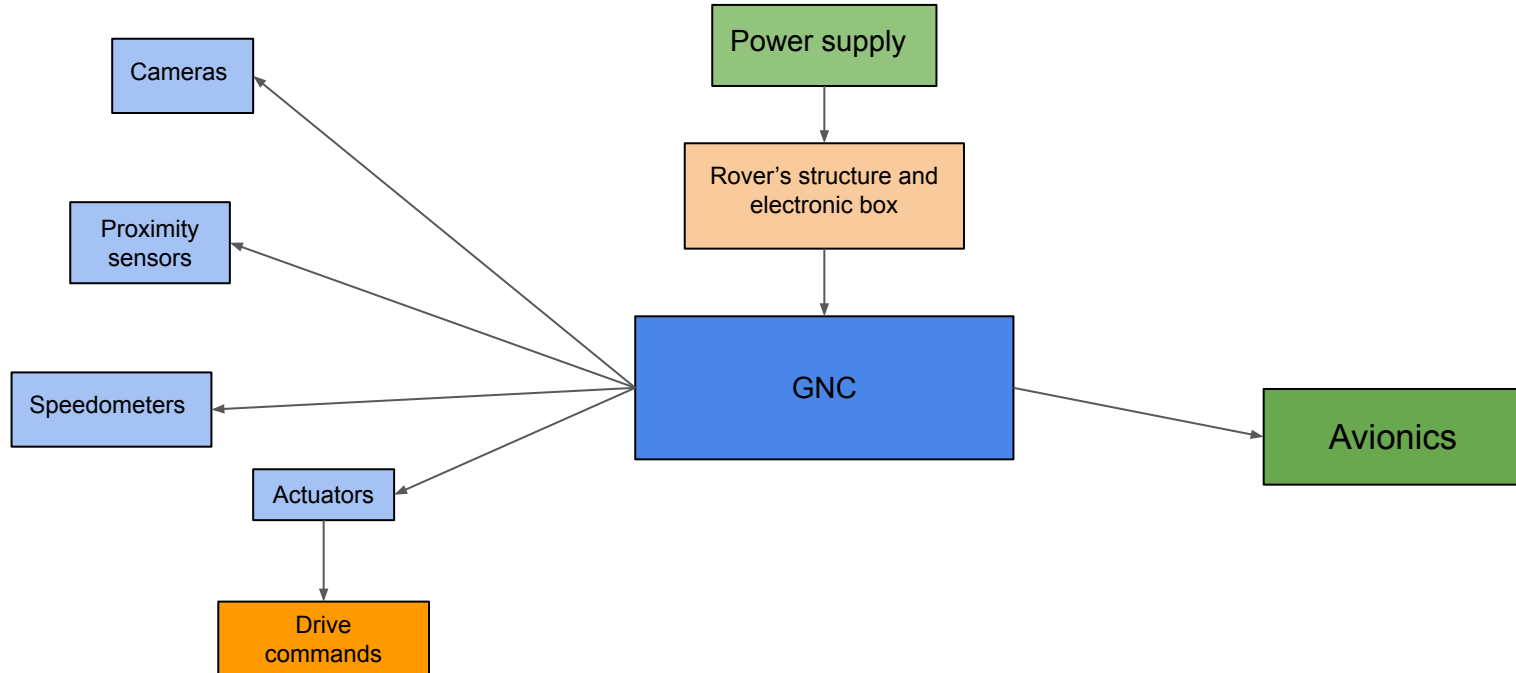
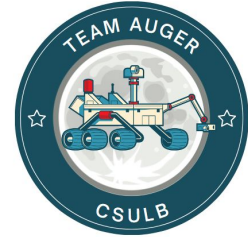
Purpose: The proposed GNC subsystem will be responsible for providing a safe and optimal route through the lunar regolith.

Requirement #	Requirement	Verification Method	Validation Statement
R.2.4.1	The rover shall traverse the lunar surface at a maximum speed of 0.6 km per hour (TBR).	Test	Provide sufficient time for the rover to properly scan the lunar regolith while looking for water.
R.2.4.2	The rover shall scan for obstacles on the lunar surface within 5 m $\pm$ 0.1 m range (TBR).	Test	To provide sufficient time for the rover to avoid the obstacle.
R.2.4.3	The rover shall maintain stability on lunar regolith with slopes of 30 degrees $\pm$ 5 degrees (TBR).	Test	To prevent the rover from falling on uneven terrains.
R.2.4.4	The rover shall respond to control inputs with a maximum delay of 500 milliseconds. (TBR)	Test	Ensuring that the rover will respond to a command promptly to avoid hazards or change direction.



GNC

Proposed Design Interfaces





GNC Subsystem Trade Studies



Steering Actuators			ENTER SCORE ↓	Electric	Pneumatic
CRITERIA	MANDATORY Y=1 / N=0	WEIGHT	SCALE		
Precision/Accuracy	1	40%	3 = Most Accurate 1 = Least Accurate	3	1
Power Efficiency	1	30%	3 = Most Efficient 1 = Least Efficient	3	1
Weight	1	15%	3 = Least Weight 1 = Most Weight	3	2
Size	0	15%	3 = Most Compact 1 = Least Compact	2	2
WEIGHTED TOTALS		100%		92%	50%



GNC Subsystem Trade Studies



Obstacle Detection System			ENTER SCORE ↓	Ginger II Radar (Radars)	URM06-PULSE (Ultrasonic Sensor)	Laser Range Finder
CRITERIA	MANDATORY Y=1 / N=0	WEIGHT	SCALE			
Range (m)	1	40%	3= Most Distance 1= Least Distance	2	1	3
Power (V)	1	30%	3 = Most Efficient 1 = Least Efficient	1	2	3
Mass (Kg)	1	10%	3= Least Weight 1= Most Weight	1	3	2
Frequency (Hz)	0	10%	3= Higher 1= Lower	3	1	2
WEIGHTED TOTALS		100%		58%	58%	84%



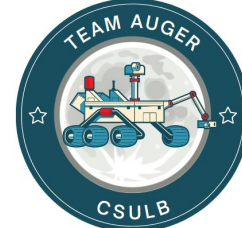
GNC Subsystem Trade Studies



Inertial Measurement System			ENTER SCORE ↓	LN-200	VN-100	Microstrain 3DM-GX5
CRITERIA	MANDATORY Y=1 / N=0	WEIGHT	SCALE			
Power (V)	1	50%	3= Most Efficient 1= Less Efficient	3	2	1
Size	1	25%	3 = Most Compact 1 = Least Compact	1	2	3
Mass	1	15%	3= Least Weight 1= Most Weight	1	3	2
Cost	0	10%	3= Least Cost 1= Most Cost	1	3	2
WEIGHTED TOTALS		100%		50%	83%	67%



Proposed Power Requirements

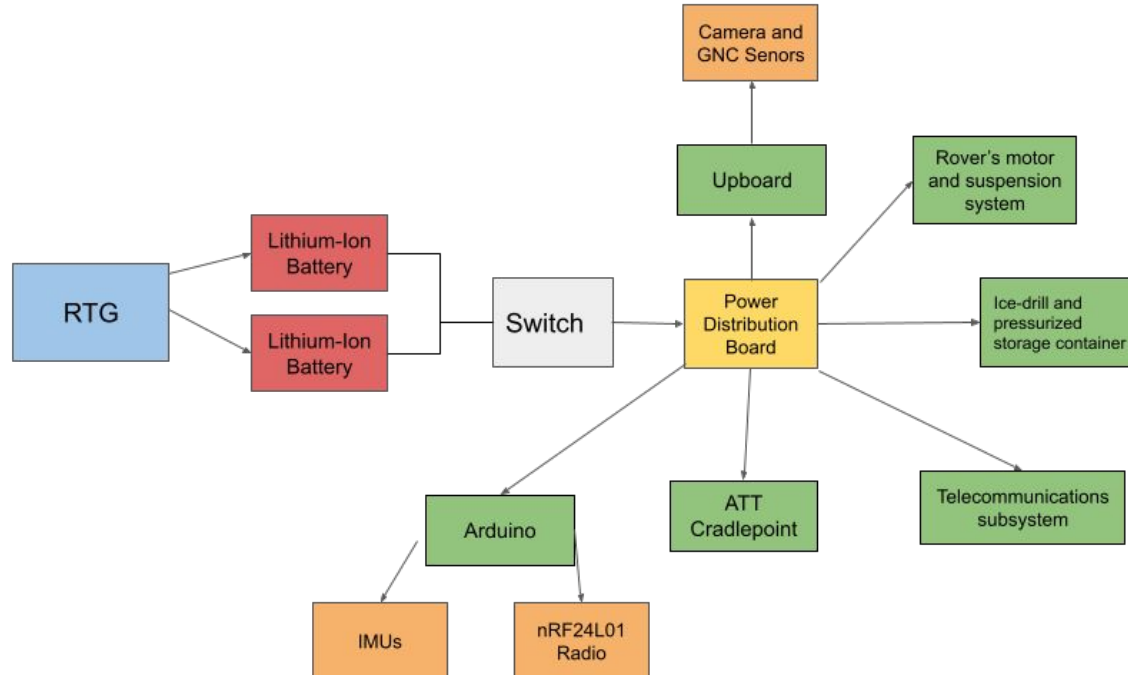
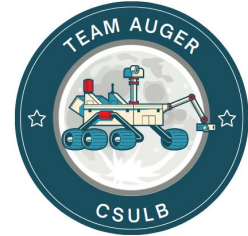


Purpose: The proposed power subsystem will provide power to all other subsystem during all operational modes.

Requirement #	Requirement Description	Verification Method	Validation Statement
R.2.5.1	The power system shall generate a minimum of 100 Watts $\pm$ 10 Watts. (TBR)	Test	Generate continuous power.
R.2.5.2	The power system shall charge 2, 450 Watt $\pm$ 10 batteries. (TBR)	Test	Store power when the demand temporarily exceeds RTG steady electrical output levels.
R.2.5.3	The power system shall provide 100 Watts $\pm$ 10 to power the avionics system. (TBR)	Test	Provide power to camera and water differentiating sensor
R.2.5.4	The power system shall provide 400 Watts $\pm$ 10 power the ice-extraction system. (TBR)	Test	Ensure that the drill and pressurized storage unit is fully operational
R.2.5.5	The power system shall provide 250 Watts $\pm$ 10 to power the movement of the rover. (TBR)	Test	Deliver continuous power to the motors in the propulsion system
R.2.5.6	The power system shall protect instrumentation from power overload with an overload relay. (TBR)	Test	Ensure that all instrumentations are protected from overcurrent and overvoltage



Power Subsystem Proposed Interfaces





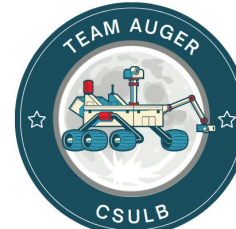
Power Subsystem Proposed Trade Studies



Radioisotope Thermoelectric Generators			ENTER SCORE ↓	Multi-Mission - RTG (MMRTG)	General-Purpose Heat Source - RTG (GPHS-RTG)	System for Nuclear Auxiliary Power (SNAP-27)
CRITERIA	MANDATORY Y=1 / N=0	WEIGHT	SCALE			
Power Output (Watts)	1	40%	3 = Most Power Output 1 = Least Power Output	2	3	1
Heat Generation (°C)	1	25%	3 = Least Heat Generated 1 = Most Heat Generated	2	1	3
Mass (kg)	1	25%	3 = Least heavy 1 = Most heavy	2	3	1
Cost	0	10%	3= Least Expensive 1= Most expensive	3	2	1
WEIGHTED TOTALS		100%		70%	80%	50%



Power Subsystem Proposed Trade Studies



Lithium-Ion Batteries			ENTER SCORE ↓	Lithium Cobalt Oxide (LCO)	Lithium Nickel Manganese Cobalt Oxide (NMC)	Lithium Nickel Cobalt Aluminum Oxide (NCA)
CRITERIA	MANDATORY Y=1 / N=0	WEIGHT	SCALE			
Capacity (Whr)	1	40%	3 = Most Power Output 1 = Least Power Output	2	1	3
Charge Rate	1	25%	3 = Largest Charge Rate 1 = Smallest Charge Rate	3	3	3
Thermal Runaway (°C)	1	25%	3 = Least heavy 1 = Most heavy	2	3	2
Cycle Life	1	10%	3= Longest Cycle Life 1= Shortest Cycle Life	2	3	1
WEIGHTED TOTALS		100%		75%	73%	85%



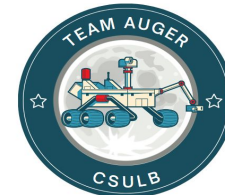
Power Subsystem Proposed Trade Studies



Power Distribution System			ENTER SCORE ↓	Radial	Loop	Network
CRITERIA	MANDATORY Y=1 / N=0	WEIGHT	SCALE			
Efficiency	1	20%	3 = Most Efficient 1 = Least Efficient	1	2	3
Reliability	1	70%	3 = Most Reliable 1 = Least Reliable	1	2	3
Cost	0	10%	3 = Least expensive 1 = Most Expensive	3	2	1
WEIGHTED TOTALS		100%		40%	67%	93%



Proposed Thermal Subsystem Requirements

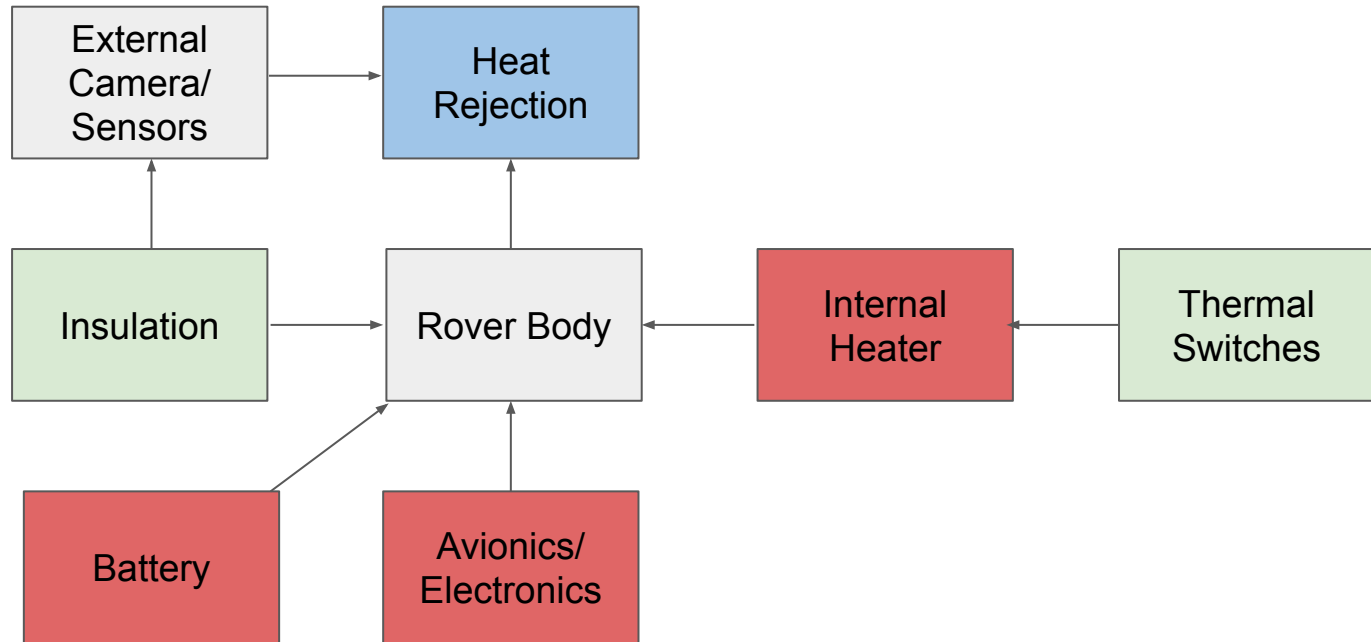
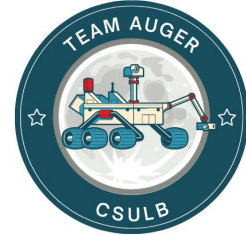


Purpose: The proposed thermal system will enable the rover to endure the extreme temperatures on the lunar south pole and in its craters, while containing and redirecting the rover's heat as well as preventing sublimation of the collected water ice stored within.

Requirement Number	Requirement	Verification Method	Validation
R.2.6.1	The rover shall operate within a temperature range of -205°C to 55°C (TBR).	Test	To meet the expected conditions during operations within the crater
R.2.6.2	The operational temperature of all internal electronic components shall be maintained between -40°C and 40°C (TBR).	Test	To prevent overheating, freezing, and malfunction of the internal electronics/avionics box
R.2.6.3	The rover's external cameras and sensors shall be maintained within the operational temperature range of -40°C and 70°C (TBR).	Test	To prevent overheating, freezing, and malfunction of external GNC/avionics components
R.2.6.4	The thermal subsystem shall maintain the temperature of the collected water in the pressurized storage tank within a range of 0°C and 100°C (TBR)	Test	To maintain the collected ice water in a liquid phase



Thermal Proposed Design Interface





Thermal Proposed Trade Studies



Heaters			ENTER SCORE ↓	Radioisotope Heat Units	Patch/Kapton/Polyimide Heater	Electrical Heater
CRITERIA	MANDATORY Y=1 / N=0	WEIGHT	SCALE			
Temperature Control	1	25%	3 = Most Precise 1 = Least Precise	3	2	3
Sustainability	1	20%	3 = Most Optimal 1 = Least Optimal	3	3	2
Energy Efficiency	1	25%	3 = Most Efficient 1 = Least Efficient	3	2	2
Sizing/Weight	1	10%	3 = Most Optimal 1 = Least Optimal	3	3	2
Temperature Uniformity	1	20%	3 = Most Uniform 1 = Least Uniform	2	2	3
WEIGHTED TOTALS		100%		94%	77%	82%



Thermal Proposed Trade Studies



Heat Rejection/Cooling Methods			ENTER SCORE ↓	Radiator	Thermoelectric Coolers	Fluid Loop System
CRITERIA	MANDATORY Y=1 / N=0	WEIGHT	SCALE			
Dissipation Rate	1	25%	3 = Most Optimal 1 = Least Optimal	3	2	3
Sustainability	1	20%	3 = Most Sustainable 1 = Least Sustainable	3	2	3
Power Usage	1	20%	3 = Most Efficient 1 = Least Efficient	3	2	2
Heating/Cooling Capacity	1	25%	3 = Most Effective 1 = Least Effective	3	2	3
Cost	0	10%	3 = Least Expensive 1 = Most Expensive	2	2	2
WEIGHTED TOTALS		100%		97%	67%	90%



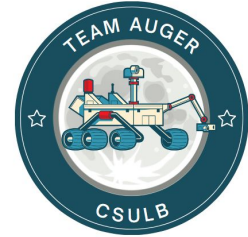
Thermal Proposed Trade Studies



Insulators			ENTER SCORE ↓	Aerogel	Foam	Multi-Layer Insulation (MLI)
CRITERIA	MANDATORY Y=1 / N=0	WEIGHT	SCALE			
Thermal Resistance	1	30%	3 = Most Effective 1 = Least Effective	3	3	3
Thermal Conductivity	1	30%	3 = Least Conductive 1 = Most Conductive	3	2	3
Density	1	15%	3 = Most Optimal 1 = Least Optimal	3	2	2
Size/Weight	0	15%	3 = Most Optimal 1 = Least Optimal	3	2	2
Cost	0	10%	3 = Least Expensive 1 = Most Expensive	2	3	2
WEIGHTED TOTALS		100%		97%	80%	85%



Proposed Telecommunication Requirements

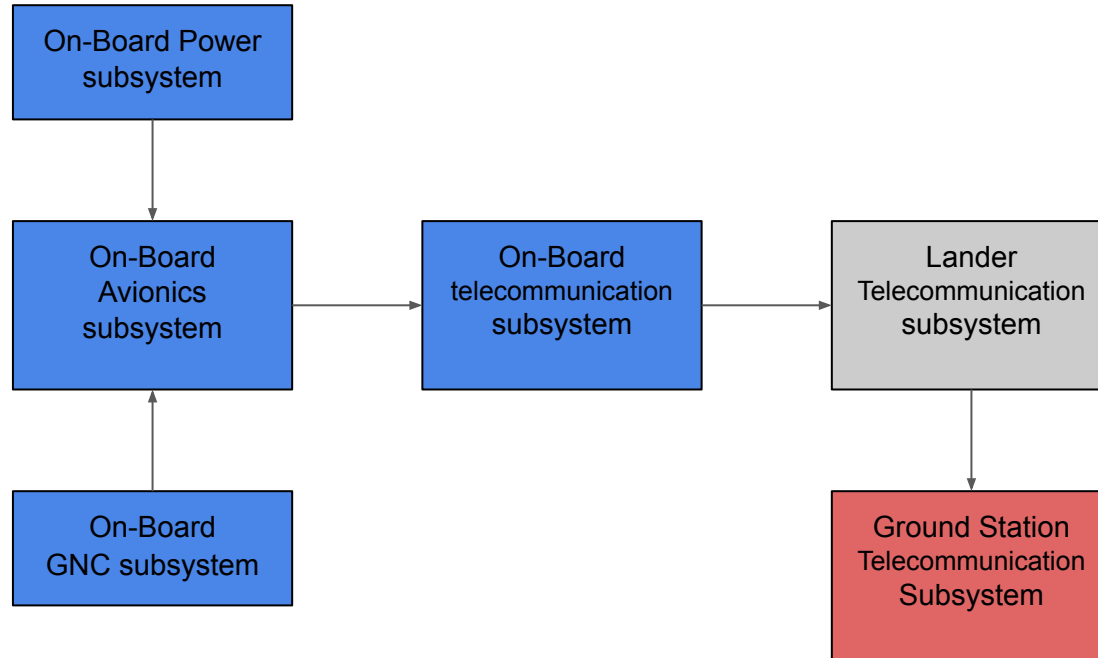
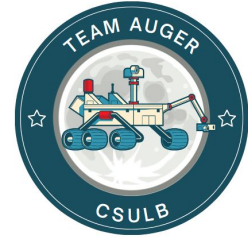


Purpose: The proposed telecommunication subsystem is responsible of transmitting packaged data collected from other subsystems and receiving data from ground stations

Requirement #	Requirement	Verification Method	Validation Statement
R.2.7.1	The telecommunication system shall operate for 3 hours ± 10 minutes per day (TBR)	Test	Provides ongoing communication from the rover to ground stations
R.2.7.2	The telecommunication subsystem frequency from the rover to the lander will range from 2 GHz to 4 GHz	Test	Provides stable communication between the rover and the lander
R.2.7.3	The telecommunication subsystem frequency from the lander to ground stations will range from 8 GHz to 12 GHz	Test	Provides communication between the lander and ground stations
R.2.7.4	The lander shall relay data transmissions at a rate of 2 Mbps ± 0.5 Mbps to ground stations	Test	Ensures successful exchange of information between the lander and ground stations

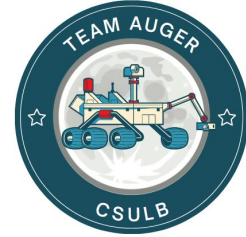


Telecommunication Proposed Design Interfaces





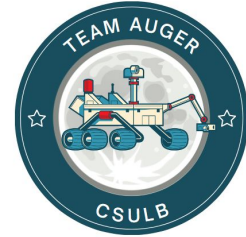
Telecommunication Subsystem Trade Studies



Radio Frequency (Rover to Lander)			ENTER SCORES	X-Band	S-Band	Ka-Band
Criteria	Mandatory Y=1/N=0	Weight	SCALE			
Data-transfer rate	1	30%	3 = most efficient 1 = least efficient	2	1	3
Power consumption	1	25%	3 = least consumption 1 = most consumption	2	3	1
Suitable for short distances	1	25%	3 = least suitable 1 = most suitable	1	3	1
Cost	0	10%	3 = least expensive 1 = most expensive	2	3	1
Uninterrupted Data Collection		100%		52%	70%	50%



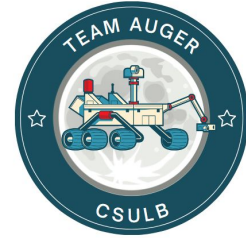
Telecommunication Subsystem Trade Studies



Radio Frequency (Lander to Ground Bases)			ENTER SCORES	X-Band	S-Band	Ka-Band
Criteria	Mandatory Y=1/N=0	Weight	SCALE			
Data-transfer rate	1	30%	3 = most efficient 1 = least efficient	2	1	3
Power consumption	1	25%	3 = least consumption 1 = most consumption	2	3	1
Suitable for long distances	1	25%	3 = least suitable 1 = most suitable	1	3	1
Cost	0	20%	3 = least expensive 1 = most expensive	2	3	1
Uninterrupted Data Collection		100%		75%	64%	70%



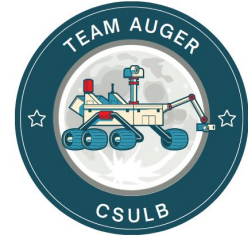
Telecommunication Subsystem Trade Studies



Antenna type		ENTER SCORES	High-Gain	Phased Array	Patch Array
Criteria	Weight	SCALE			
Communication Range	35%	3 = smallest range 1 = largest range	2	1	3
Power consumption	25%	3 = least consumption 1 = most consumption	2	3	1
Weight	15%	3 = least heavy 1 = most heavy	2	3	3
Lifespan	15%	3 = least durable 1 = most durable	1	2	2
cost	10%	3 = least expensive 1 = most expensive	3	1	2
Uninterrupted Data Collection	100%		65%	62%	47%



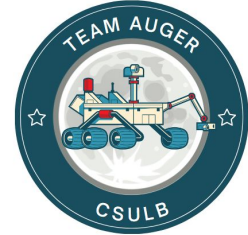
Issues & Concerns



Our missions main concern would be to withstand the cold conditions in the permanently shadowed area, which is located at the South Pole of the Moon to investigate the concentration of water ice.



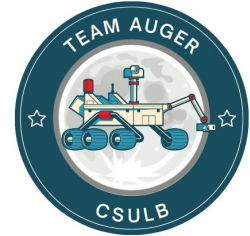
Team Leader Assessment



I have confidence in Team Auger's capabilities to design and develop our rover and meet our mission requirements. While we acknowledge the potential risks and challenges that may arise, we are committed to addressing these by continuously reviewing our design, conducting in-depth research, rigorous testing, and maintaining an open-minded approach. Through these efforts, we are assured for success in creating a rover capable of effectively searching, drilling, and storing ice water at the South Pole of the Moon.



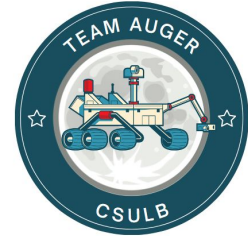
Q & A



Thank you, any questions or concerns?



References



Navigation Mars and Beyond.

<https://www.northropgrumman.com/space/navigating-mars-and-beyond-with-ln-200-inertial-measurement-units> Accessed October 20th, 2023.

LN-200 FOG Family Advanced Airborne IMU/AHRS.

<https://www.northropgrumman.com/what-we-do/air/ln-200-fog-family-advanced-airborne-imu-ahrs> Accessed October 20th, 2023

Sensor Fusion In Autonomous Mobile Robots.

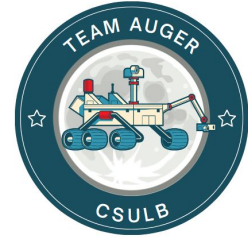
<https://www.einfochips.com/blog/sensor-fusion-in-autonomous-mobile-robots-enabling-amrs-to-navigate-safely-and-precisely/#:~:text=With%20sensor%20fusion%2C%20accurate%20path.and%20keeps%20on%20operating%20seamlessly>. Accessed October 20th, 2023.

Design and field testing of a rover with an actively articulated suspension system in a Mars analog terrain

<https://onlinelibrary.wiley.com/doi/10.1002/rob.21808> Accessed October 17th, 2023



References



5.0 Guidance, Navigation and Control.

[https://www.nasa.gov/smallsat-institute/sst-soa/guidance-navigation-and-control/#:~:text=The%20Guidance%2C%20Navigation%20%26%20Control%20,Positioning%20System%20\(GPS\)%20receiver](https://www.nasa.gov/smallsat-institute/sst-soa/guidance-navigation-and-control/#:~:text=The%20Guidance%2C%20Navigation%20%26%20Control%20,Positioning%20System%20(GPS)%20receiver). Accessed October 19th, 2023.

Guidance, Navigation and Control Systems.

<https://mars.nasa.gov/mro/mission/spacecraft/parts/gnc/> Accessed October 19th, 2023.

System Modelling of Rocker-Bogie Mechanism for Disaster Relief. *Procedia Computer Science*, Elsevier, 29 Dec. 2015,

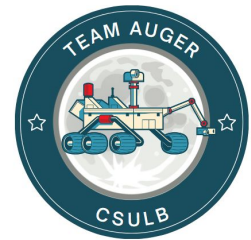
www.sciencedirect.com/science/article/pii/S1877050915038508?ref=pdf_download&fr=RR-2&rr=81a753f7884e2aa4. Accessed October 18th, 2023

DFROBOT Drive the Future.

<https://www.dfrobot.com/product-2059.html> Accessed October 19th, 2023.



References



"7.0 Thermal Control." NASA, NASA, 28 Sept. 2023,

www.nasa.gov/smallsat-institute/sst-soa/thermal-control/#7.2.7.

"The Rover's Temperature Controls." NASA, NASA, Accessed 20 Oct. 2023.

Battery University. "BU-205: Types of Lithium-Ion." Battery University, 18 Oct. 2022,

batteryuniversity.com/article/bu-205-types-of-lithium-ion.

mars.nasa.gov/mer/mission/rover/temperature/

"Radioisotope Thermoelectric Generator." *Wikipedia*, Wikimedia Foundation, 6 Oct. 2023,

en.wikipedia.org/wiki/Radioisotope_thermoelectric_generator.

"Ice on the Moon." NASA, NASA, Accessed 19 Oct. 2023.

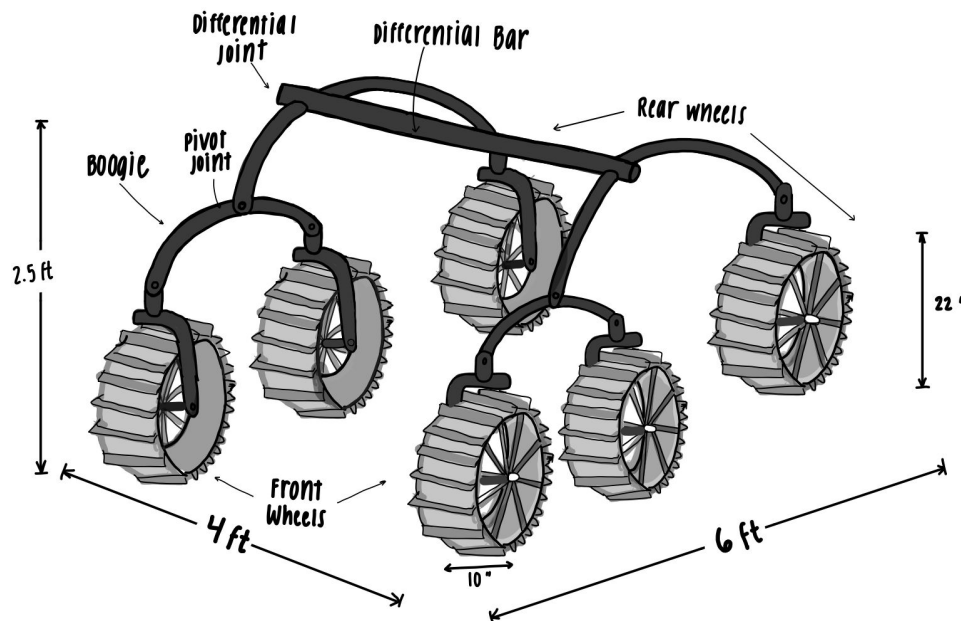
nssdc.gsfc.nasa.gov/planetary/ice/ice_moon.html

"The Moonranger Story." *MoonRanger*, 15 Nov. 2021,

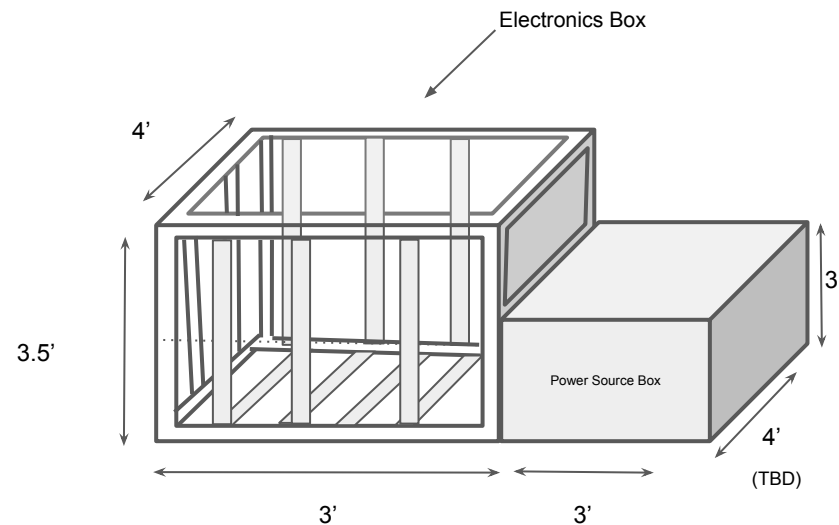
labs.ri.cmu.edu/moonranger/blog/



Preliminary Structure Design



Proposed Suspension System Design



Rover Chassis Frame



Power Subsystem Proposed Trade Studies



Power Distribution System			ENTER SCORE ↓	Regulated	Unregulated
CRITERIA	MANDATORY Y=1 / N=0	WEIGHT	SCALE		
Efficiency	1	25%	3 = Most Efficient 1 = Least Efficient	3	1
Weight	1	25%	3 = Least Heavy 1 = Most Heavy	2	3
Precision	1	25%	3 = Most Precise 1 = Least Precise	3	1
Cost	0	25%	3 = Least expensive 1 = Most Expensive	1	3
WEIGHTED TOTALS		100%		75%	67%